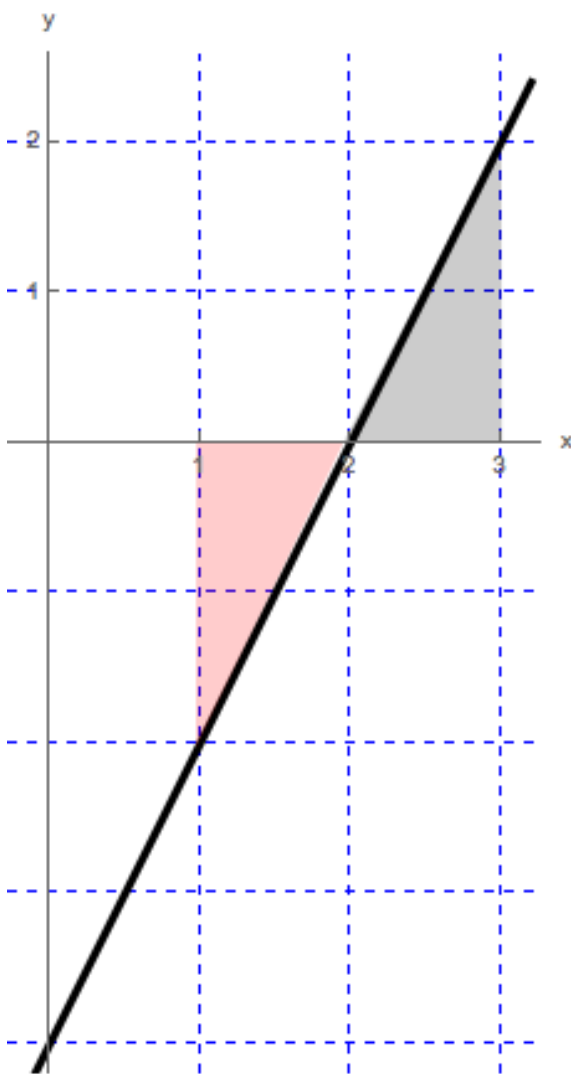


Problem 4

Problem 4

Problem 1 Use geometry to evaluate the definite integral. Sketch the graph of the function and shade the relevant regions.

(a) $\int_1^3 (2x - 4) dx$



Explanation.

Problem 4

We want to find the net area of the shaded region. We have two identical triangles, each with base 1 and height 2. Since the triangles will have identical area and one lies above and the other below the x -axis, the total area will be 0.

Therefore, $\int_1^3 (2x - 4) dx = 0$

(b) $\int_1^3 |2x - 4| dx$

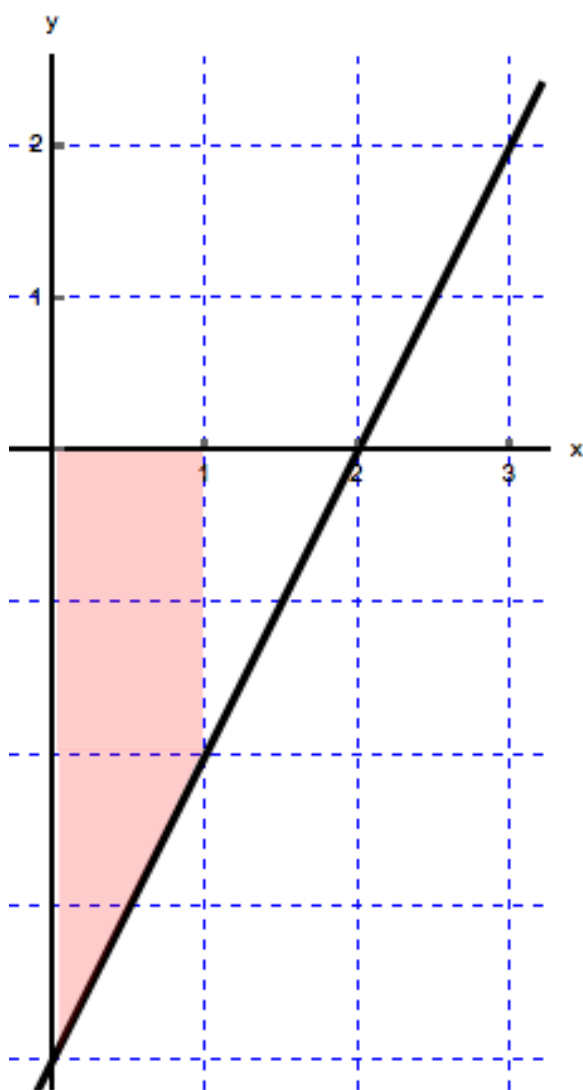
Explanation. The function changes the sign at $x = 2$. Therefore, we will split the integral into two integrals:

$$\int_1^3 |2x - 4| dx = \int_1^2 |2x - 4| dx + \int_2^3 |2x - 4| dx.$$

Since the function is negative for $x < 2$ and positive for $x > 2$, we write

$$\int_1^3 |2x - 4| dx = - \int_1^2 (2x - 4) dx + \int_2^3 (2x - 4) dx = 1 + 1 = 2.$$

(c) $\int_0^1 (2x - 4) dx$



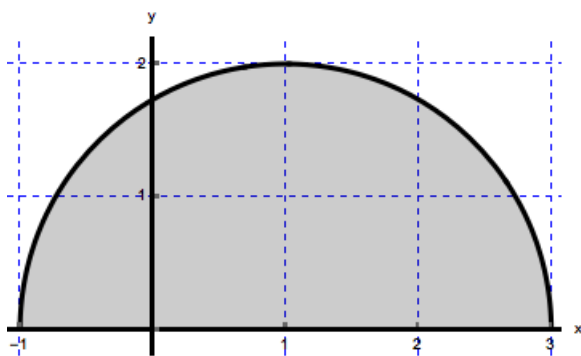
Explanation.

Here we need to find the area of a trapezoid. The trapezoid is made up of a rectangular top with area $(2)(1) = 2$ and triangle bottom with area $(1/2)(1)(2) = 1$. This gives us an area of 3, but, since f is negative on the interval $(0, 1)$, the value of the integral is the negative of this area.

Therefore, $\int_0^1 (2x - 4) dx = -3$

(d) $\int_{-1}^3 \sqrt{4 - (x - 1)^2} dx$

Problem 4



Explanation.

We are looking for the area of the entire semi-circle above the x -axis. This is a circle with radius 2 so we have $(1/2)\pi 2^2 = 2\pi$

$$\int_{-1}^3 \sqrt{4 - (x - 1)^2} dx = 2\pi$$